# EPSR-test-ex49 readpdb: single-atom residues

Program(s): readpdb | chained passes: 1 | verification: ALL IDENTICAL (1 pass)

## What this case tests

Exercises `readpdb` [1,2] with residues of a single atom (ions), element names taken from columns 13-14 of the atom name with upper-case symbol matching (`NA`, `CL`), and a `HETATM` record, which `readpdb` ignores because only `ATOM` records are read.

## Code exercised

**Fortran (original, fortran_src/):**

- `readpdb.f`: `run_readpdb`, `readpdb` (CRYST1/ATOM/CONECT records, molecules from residues, bonds from van der Waals radii, angle constraints, DIH groups), `upcase`
- `ato_manipulation_rev4.f`: `ctype`, `ftype`, `centreofmass`, `get_rotgrp`, `dihedral`
- `auxiliary_routines.f`: `convertalphabetagamma`; `ato_read_and_save.f`: `moloutc`

**Rust (port, rust/epsr-core/src/):**

- `tools/readpdb.rs`: `run`, `readpdb`, `conect` (molecule merging), `make_dihedrals`, `read_fixed`, `upcase`
- `tools/readxyz.rs`: `set_type_parameters`; `ato.rs`: `State::ctype`, `State::ftype`
- `tools/groupc_common.rs`: `get_rotgrp`, `dihedral`, `convertalphabetagamma`, `setup_lattice`, `centreofmass`, `moloutc` (with `DIH` groups)

## Physical system

A hand-made aqueous NaCl fragment in a cubic 15 Å cell: three water molecules, one Na and one Cl residue (5 molecules, 11 atoms read; a `HETATM` carbon is skipped). Bonds are found where the distance is below 0.75 × the sum of the van der Waals radii; no dihedral groups.

## Inputs

`ions.pdb` written by hand; stdin `readpdb.stdin`: factor `0.75`, dihedrals `n`; `vanderWaalsRadii.txt` copied from EPSR-test.

## How the accuracy was verified

`commands.txt` (1 program call, run once) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; no masking except where noted below). The Fortran output is stored in `fortran_reference.tar.gz`. `readxyz`/`readpdb` crash (SIGSEGV) as compiled from the original source, so the reference build adds an explicit INTERFACE block for `ctype` (`rust/tools/build_patched_readxyz_readpdb.sh`); the port behaves like the patched program. Result of the final regression run of the complete port: every file identical in 1 pass.
Output compared: `ions.ato`, in which the Na and Cl molecules appear with `nato` = `1` but no atom records and their centres at the origin, exactly as written by the Fortran. Fortran reference: the original `readxyz`/`readpdb` binaries crash (SIGSEGV) at the first call of `ctype`, which has optional arguments but is called with 11 of its 13 arguments without an explicit interface (gfortran passes the hidden character lengths where `radmaxsph`/`pseudofacsph` are expected). The reference was therefore produced with an explicit `INTERFACE` block for `ctype` added to the program (no other change; absent arguments mean `radmaxsphere` = `0.5*sigcor`, `pseudofacmol` = `1`), built by `rust/tools/build_patched_readxyz_readpdb.sh`; the comparison is against this patched build, not the original binary.

## Notes

Fortran quirk reproduced: for a one-atom residue `nal` is never set (it is assigned only when a second atom is added) and keeps 0, so the loops `naf..nal` are empty and the ions are effectively lost.

## References


[1] wwPDB, "Atomic Coordinate Entry Format Description", version 3.3 (2012), https://www.wwpdb.org/documentation/file-format.
[2] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).
[3] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.